V1 Enterprise Authentication Dashboard

Power BI Build Guide — Complete DAX, Visuals & Layout

Data Source: Hive Data Lake (2 tables)

v1_customer_dim — 14.2M rows — customer demographics, digital status, FI status

v1_session_events — 1.17B rows — all authentication & session events

Database: dm_ib_dev

Dashboard Pages:

- Page 1: Executive Overview & KPIs
- Page 2: OTP & MFA Deep Dive
- Page 3: Session Events, SSO & FI Status

Requirements Covered: 19 (all V1 items)

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1. Data Model Setup

1.1 Import Tables

In Power BI Desktop, go to **Get Data** → **Hive LLAP** (or your Hive connector). Import:

- **dm_ib_dev.v1_customer_dim** — dimension table
- **dm_ib_dev.v1_session_events** — fact table

1.2 Create Relationship

In **Model View**, drag **v1_session_events.user_id** onto **v1_customer_dim.RCIF_CUST_NBR**.

Setting	Value
Cardinality	Many-to-One (events → customer)
Cross-filter	Single direction
Active	Yes

1.3 Column Mapping Reference

Table	Key Columns	Purpose
v1_customer_dim	RCIF_CUST_NBR, AGE_BANDING, CUSTOMER_GENERATION, DIGITAL_CUSTOMER_CHECK, FI_STATUS	Customer dimension for slicers and grouping
v1_session_events	user_id, source_table, action, authentication_method, category, session_type, event_date	Fact table with all auth events

2. Date Table

Create a calculated table for time intelligence. In Power BI Desktop: **Modeling** → **New Table**:

```
DateTable =
ADDCOLUMNS(
CALENDARAUTO(),
"Year", YEAR([Date]),
"Month", FORMAT([Date], "MMMM"),
"MonthNum", MONTH([Date]),
"Quarter", "Q" &amp; FORMAT([Date], "Q"),
"Week", WEEKNUM([Date]),
"Day", DAY([Date]),
"Weekday", FORMAT([Date], "dddd")
)
```

Then: right-click **DateTable** → **Mark as date table** → select the **Date** column.

Relationship: Drag **v1_session_events.event_date** → **DateTable.Date** (Many-to-One).

If event_date is a string, first convert: Go to Power Query → select event_date → Transform → Data Type → Date.

3. DAX Measures — All 19 Requirements

Create all measures in a dedicated **_Measures** table (Home → Enter Data → name it _Measures → Load). Then create each measure below inside that table.

Req 2 — Channel Type

Shows distinct channel types. **Visual:** Use session_type as a row/axis dimension directly.

```
Channel Types =
CALCULATE(
DISTINCTCOUNT(v1_session_events[session_type]),
v1_session_events[source_table] = "SESSION"
)
```

Req 3 — Channel Code

```
Channel Codes =
CALCULATE(
DISTINCTCOUNT(v1_session_events[application_id]),
v1_session_events[source_table] = "SESSION"
)
```

Req 4 — Channel Session ID

No measure needed. Use **session_id** in table/matrix visuals for drill-through detail.

Req 5 — Device / Platform

Use **device_os_type**, **device_os_version**, **device_model** from DEVICE source as dimensions.

```
Device Users =
CALCULATE(
DISTINCTCOUNT(v1_session_events[user_id]),
v1_session_events[source_table] = "DEVICE"
)
```

Req 6 — Age Groups

No measure needed. Use **v1_customer_dim.AGE_BANDING** as slicer or axis on any visual.

Req 7 — Digitally Enrolled

```
Digital Customers =
CALCULATE(
DISTINCTCOUNT(v1_customer_dim[RCIF_CUST_NBR]),
v1_customer_dim[DIGITAL_CUSTOMER_CHECK] = "Digital Customer"
)
```

```
Non-Digital Customers =
CALCULATE(
DISTINCTCOUNT(v1_customer_dim[RCIF_CUST_NBR]),
v1_customer_dim[DIGITAL_CUSTOMER_CHECK] = "Non-Digital Customer"
)
```

Req 8 — Digitally Active Users

Users with at least one session event in the reporting period.

```
Digitally Active Users =
CALCULATE(
DISTINCTCOUNT(v1_session_events[user_id]),
v1_session_events[source_table] = "SESSION"
)
```

Req 9 — SMS OTP Users

```
SMS OTP Users =
CALCULATE(
DISTINCTCOUNT(v1_session_events[user_id]),
v1_session_events[source_table] = "AUTH",
v1_session_events[authentication_method] = "otp",
v1_session_events[category] = "sms"
)
```

```
SMS OTP Events =
CALCULATE(
COUNTROWS(v1_session_events),
v1_session_events[source_table] = "AUTH",
v1_session_events[authentication_method] = "otp",
v1_session_events[category] = "sms"
)
```

Req 10 — Email OTP Users

```
Email OTP Users =
CALCULATE(
DISTINCTCOUNT(v1_session_events[user_id]),
v1_session_events[source_table] = "AUTH",
v1_session_events[authentication_method] = "otp",
v1_session_events[category] = "email"
)
```

Req 11 — Voice Call MFA Users

From PINDROP table. Counts users with authentication_result = 'PASS'.

```
Voice MFA Users =
CALCULATE(
DISTINCTCOUNT(v1_session_events[user_id]),
v1_session_events[source_table] = "PINDROP",
v1_session_events[authentication_result] = "PASS"
)
```

```
Voice MFA Total =
CALCULATE(
DISTINCTCOUNT(v1_session_events[user_id]),
v1_session_events[source_table] = "PINDROP"
)
```

Req 12 — % UID/PWD Authentication

```
Total Auth Assertions =
CALCULATE(
COUNTROWS(v1_session_events),
v1_session_events[source_table] = "AUTH",
v1_session_events[action] = "assertion_end"
)
```

```
UID PWD Events =
CALCULATE(
COUNTROWS(v1_session_events),
v1_session_events[source_table] = "AUTH",
v1_session_events[authentication_method] = "password",
v1_session_events[action] = "assertion_end"
)
```

```
UID PWD Pct =
DIVIDE([UID PWD Events], [Total Auth Assertions], 0)
```

Format as percentage in the visual: select measure → Format → Percentage.

Req 13 — Logout Events

```
Logout Events =
CALCULATE(
COUNTROWS(v1_session_events),
v1_session_events[source_table] = "ENRICHMENT",
v1_session_events[action] = "session_logout"
)
```

Req 14 — Token Refresh Events

```
Token Refresh Events =
CALCULATE(
COUNTROWS(v1_session_events),
v1_session_events[source_table] = "ENRICHMENT",
v1_session_events[action] = "token_refresh"
)
```

Req 15 — Event Timestamp

No measure needed. Use **event_time** column in table visuals for row-level timestamps. The **event_date** column is for date-level aggregation on charts.

Req 16 — Passwordless Auth %

```
Passwordless Events =
CALCULATE(
COUNTROWS(v1_session_events),
v1_session_events[source_table] = "AUTH",
v1_session_events[action] = "assertion_end",
v1_session_events[authentication_method] &lt;&gt; "password"
)
```

```
Passwordless Pct =
DIVIDE([Passwordless Events], [Total Auth Assertions], 0)
```

Req 17 — FI Enabled vs Disabled

This is now a column on **v1_customer_dim.FI_STATUS**. Use as slicer or axis.

```
FI Enabled Customers =
CALCULATE(
DISTINCTCOUNT(v1_customer_dim[RCIF_CUST_NBR]),
v1_customer_dim[FI_STATUS] = "FI Enabled"
)
```

```
FI Disabled Customers =
CALCULATE(
DISTINCTCOUNT(v1_customer_dim[RCIF_CUST_NBR]),
v1_customer_dim[FI_STATUS] = "FI Disabled"
)
```

Req 18 — SSO / Aggregators

SSO sessions are identified by flow_id, token_name, application_id from SESSION source. Refine the filter values once you inspect actual data.

```
SSO Sessions =
CALCULATE(
COUNTROWS(v1_session_events),
v1_session_events[source_table] = "SESSION",
NOT ISBLANK(v1_session_events[token_name])
)
```

```
SSO Unique Users =
CALCULATE(
DISTINCTCOUNT(v1_session_events[user_id]),
v1_session_events[source_table] = "SESSION",
NOT ISBLANK(v1_session_events[token_name])
)
```

Req 19 — MFA Type Breakdown

Use **authentication_method** and **category** as dimensions in a matrix visual.

```
MFA Events =
CALCULATE(
COUNTROWS(v1_session_events),
v1_session_events[source_table] = "AUTH",
v1_session_events[action] = "assertion_end"
)
```

Place authentication_method on Rows, category on Columns, MFA Events as Value in a Matrix visual.

4. Page 1: Executive Overview & KPIs

4.1 Layout

This page gives stakeholders the headline numbers at a glance.

Position	Visual Type	Content
Top row (6 cards)	Card	Digital Customers | Digitally Active | UID/PWD % | Passwordless % | FI Enabled | FI Disabled
Middle left	Clustered Bar Chart	Auth Method Breakdown Axis: authentication_method Value: [Total Auth Assertions] Filter: source=AUTH, action=assertion_end
Middle right	Donut Chart	Digital vs Non-Digital Legend: DIGITAL_CUSTOMER_CHECK Value: DISTINCTCOUNT of RCIF_CUST_NBR
Bottom left	Line Chart	Daily Auth Trend Axis: event_date Value: COUNTROWS Filter: source=AUTH
Bottom right	Stacked Bar	Age Group Breakdown Axis: AGE_BANDING Values: [SMS OTP Users], [Email OTP Users], [Voice MFA Users]

4.2 Card Visual Settings

For each card: select Card visual → drag the measure to the Fields well. Format:

- Font size: 28pt for the value
- Category label: ON (shows measure name)
- For percentages: select the measure → Column tools → Format → Percentage
- Conditional formatting: green if Passwordless % > 30%, orange otherwise

4.3 Bar Chart Detail

Clustered Bar Chart for auth method breakdown:

- **Y-axis:** authentication_method (from v1_session_events)
- **X-axis:** [Total Auth Assertions] measure
- **Visual-level filter:** source_table = AUTH, action = assertion_end
- **Data labels:** ON, show as percentage of total
- **Sort:** By X-axis descending

4.4 Line Chart Detail

Shows daily authentication volume trend:

- **X-axis:** DateTable[Date]
- **Y-axis:** COUNTROWS(v1_session_events)
- **Visual-level filter:** source_table = AUTH
- Add a trend line: Format → Analytics → Trend line → ON

5. Page 2: OTP & MFA Deep Dive

5.1 Layout

Position	Visual Type	Content
Top row (3 cards)	Card	SMS OTP Users | Email OTP Users | Voice MFA Users
Middle left	Stacked Bar Chart	OTP by Age Group Axis: AGE_BANDING Legend: category (sms/email) Value: DISTINCTCOUNT(user_id) Filter: source=AUTH, authentication_method=otp
Middle right	Matrix	MFA Type Breakdown Rows: authentication_method Columns: category Values: COUNTROWS Filter: source=AUTH, action=assertion_end
Bottom left	Clustered Bar	OTP by Generation Axis: CUSTOMER_GENERATION Legend: category Value: DISTINCTCOUNT(user_id)
Bottom right	Table	Voice MFA Detail Columns: user_id, risk_score, authentication_result, line_of_business, event_date Filter: source=PINDROP

5.2 Stacked Bar — OTP by Age Group

This is the key visual for reqs 9-10. Setup:

- **Y-axis:** v1_customer_dim[AGE_BANDING]
- **X-axis:** DISTINCTCOUNT(v1_session_events[user_id])
- **Legend:** v1_session_events[category]
- **Visual filter:** source_table = AUTH, authentication_method = otp
- Sort AGE_BANDING by custom sort order if needed (Less than 18 → 18-24 → ... → 65+)

5.3 Matrix — MFA Type

Shows all authentication methods crossed with categories:

- **Rows:** authentication_method
- **Columns:** category
- **Values:** [MFA Events] or COUNTROWS
- Enable **stepped layout** OFF for clean grid
- Enable conditional formatting: Background color scale on values

5.4 Voice MFA Table

Drill-through detail for PINDROP events:

- Add visual-level filter: source_table = PINDROP
- Show columns: user_id, authentication_result, risk_score, pindrop_device_type, line_of_business, duration, event_date
- Enable conditional formatting on risk_score: red for high risk

6. Page 3: Sessions, SSO & FI Status

6.1 Layout

Position	Visual Type	Content
Top row (4 cards)	Card	Logout Events | Token Refresh | SSO Sessions | SSO Users
Middle left	Line Chart	Logout + Token Refresh Trend Axis: event_date Values: [Logout Events], [Token Refresh Events] Two lines on same chart
Middle right	Donut Chart	FI Status Distribution Legend: FI_STATUS Value: DISTINCTCOUNT of RCIF_CUST_NBR
Bottom left	Clustered Bar	SSO by Application Axis: application_id Value: COUNTROWS Filter: source=SESSION, token_name IS NOT BLANK
Bottom right	Matrix	Channel × Device Breakdown Rows: session_type Columns: device_os_type Values: DISTINCTCOUNT(user_id)

6.2 Dual-Line Chart — Logout + Token Refresh

Shows two event types trending over time:

- **X-axis:** DateTable[Date]
- **Y-axis:** [Logout Events] and [Token Refresh Events] as two separate values
- Both measures use source_table = ENRICHMENT with their respective action filters
- Format: different colors for each line, add data labels for peaks

6.3 FI Status Donut

Simple breakdown of FI Enabled vs Disabled vs Unknown:

- **Legend:** v1_customer_dim[FI_STATUS]
- **Values:** DISTINCTCOUNT(v1_customer_dim[RCIF_CUST_NBR])
- Use conditional colors: Green = Enabled, Red = Disabled, Grey = Unknown

6.4 SSO by Application

Bar chart showing which applications have SSO/aggregator traffic:

- **Y-axis:** application_id
- **X-axis:** COUNTROWS or DISTINCTCOUNT(user_id)
- **Visual filter:** source_table = SESSION, token_name IS NOT BLANK

- Top N filter: show only top 10 applications by count

7. Slicers & Filters (All Pages)

Add these slicers to **every page** for consistent filtering. Use the Sync Slicers pane (View → Sync Slicers) to sync across all pages.

Slicer	Table.Column	Type	Position
Date Range	DateTable[Date]	Between (date range)	Top left
Age Group	v1_customer_dim[AGE_BANDING]	Dropdown	Left sidebar
Generation	v1_customer_dim[CUSTOMER_GENERATION]	Dropdown	Left sidebar
Digital Status	v1_customer_dim[DIGITAL_CUSTOMER_CHECK]	Buttons (2)	Left sidebar
FI Status	v1_customer_dim[FI_STATUS]	Buttons (3)	Left sidebar
Channel	v1_session_events[session_type]	Dropdown	Left sidebar

7.1 Slicer Tips

- Date Range slicer: Format as slider with start/end date pickers
- Use the Slicer header to show a clear label
- Set default date range to match your data window (last 3 or 6 months)
- For Age Group: set custom sort order → go to Column tools → Sort by Column → create a sort index

7.2 Age Banding Sort Order

AGE_BANDING text values don't sort correctly alphabetically. Create a calculated column:

```
Age Sort =
SWITCH(
v1_customer_dim[AGE_BANDING],
"Less than 18", 1,
"18-24", 2,
"25-34", 3,
"35-44", 4,
"45-54", 5,
"55-64", 6,
"65+", 7,
8
)
```

Then: select AGE_BANDING column → Column tools → Sort by Column → Age Sort.

8. Performance Tips

With 1.17B rows in the session events table, Power BI may struggle. Here are strategies:

8.1 Import Mode vs DirectQuery

Import Mode (recommended): Loads all data into PBI's in-memory engine. Fastest visuals but needs enough RAM. For 1.17B rows, this may require 16-32GB RAM on the PBI machine.

DirectQuery: Queries Hive on every interaction. Slower visuals but no RAM constraint. Good if your Hive cluster is fast.

Recommendation: Start with Import Mode. If RAM is an issue, create a pre-aggregated summary table.

8.2 Pre-Aggregated Summary Table (Optional)

If Import Mode is too heavy, create a third Hive table that pre-aggregates daily:

```
-- Run in CDSW as step2
SELECT
event_date,
source_table,
authentication_method,
category,
action,
session_type,
COUNT(*) AS event_count,
COUNT(DISTINCT user_id) AS unique_users
FROM dm_ib_dev.v1_session_events
GROUP BY 1,2,3,4,5,6
```

This would shrink from ~1.17B rows to maybe 50K-200K rows. Import that for most visuals, and use the detail table only for drill-through.

8.3 Other Tips

- Remove unused columns in Power Query before loading (e.g., client_ip, hostname, device_hw_id)
- Set data types correctly (event_date as Date, failure as Integer)
- Disable Auto date/time: File → Options → Data Load → uncheck Auto date/time
- Use DISTINCTCOUNT carefully — it's expensive on 1B+ rows

9. Quick Validation Queries

Run these in CDSW or Hue to cross-check PBI numbers:

9.1 Total Digital Customers

```
SELECT COUNT(DISTINCT RCIF_CUST_NBR)
FROM dm_ib_dev.v1_customer_dim
WHERE DIGITAL_CUSTOMER_CHECK = 'Digital Customer'
```

9.2 SMS OTP Unique Users

```
SELECT COUNT(DISTINCT user_id)
FROM dm_ib_dev.v1_session_events
WHERE source_table = 'AUTH'
AND authentication_method = 'otp'
AND category = 'sms'
```

9.3 UID/PWD %

```
SELECT
ROUND(
SUM(CASE WHEN authentication_method = 'password'
THEN 1 ELSE 0 END) * 100.0 /
COUNT(*), 2
) AS uid_pwd_pct
FROM dm_ib_dev.v1_session_events
WHERE source_table = 'AUTH'
AND action = 'assertion_end'
```

9.4 FI Status Counts

```
SELECT FI_STATUS, COUNT(DISTINCT RCIF_CUST_NBR)
FROM dm_ib_dev.v1_customer_dim
GROUP BY FI_STATUS
```

9.5 Event Counts by Source Table

```
SELECT source_table, COUNT(*)
FROM dm_ib_dev.v1_session_events
GROUP BY source_table
ORDER BY 2 DESC
```

9.6 Logout + Token Refresh Counts

```
SELECT action, COUNT(*)
FROM dm_ib_dev.v1_session_events
WHERE source_table = 'ENRICHMENT'
GROUP BY action
```

Match every PBI card/chart against these queries. If numbers diverge, check your visual-level filters first.